

Holonomy and singular foliations

Marco Zambon (Univ. Autónoma Madrid-ICMAT)

joint work with
Iakovos Androulidakis (University of Athens)

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Introduction

We study geometric properties of singular foliations:

- A) Is there any sense in which the holonomy groupoid of a singular foliation is **smooth**?
- B) What is the notion of **holonomy** for a singular foliation?
- C) When is a singular foliation isomorphic to its **linearization**?

For a **regular foliation** given by an involutive distribution $F \subset TM$, it is well known that:

B) Given a path $\gamma: [0, 1] \rightarrow M$ lying in a leaf, its **holonomy** is the germ of a diffeomorphism $S_{\gamma(0)} \rightarrow S_{\gamma(1)}$ between slices transverse to F . It is obtained “following nearby paths in leaves of F ”.

A) The **holonomy groupoid** is

$$H = \{\text{paths in leaves of } F\} / (\text{holonomy of paths}).$$

It is a Lie groupoid, integrating the Lie algebroid F .

C) **Non-invariant Reeb stability theorem:**

Suppose L is an embedded leaf and H_x^x is finite ($H_x^x =$ holonomy of loops based at $x \in L$).

Then, nearby L , the foliation F is isomorphic to its linearization.

Singular foliations

Let M be a manifold.

A **singular foliation** $\mathcal{F}$ is a submodule of the $C^\infty(M)$ -module $\mathcal{X}_c(M)$ (the compactly supported vector fields) such that:

- $\mathcal{F}$ is locally finitely generated,
- $[\mathcal{F}, \mathcal{F}] \subset \mathcal{F}$.

$(M, \mathcal{F})$ is partitioned into leaves (of varying dimension).

Examples

- 1) On $M = \mathbb{R}$ take $\mathcal{F}$ to be generated by $x\partial_x$ or by $x^2\partial_x$. Both foliations have the same partition into leaves: $\mathbb{R}_-, \{0\}, \mathbb{R}_+$.
- 2) On $M = \mathbb{R}^2$ take $\mathcal{F} = \langle \partial_x, y\partial_y \rangle$. •
- 3) If $B \rightarrow M$ is a Lie algebroid with anchor ρ , then

$$\mathcal{F} = \rho(\Gamma_c(B))$$

is a singular foliation.

A) The holonomy groupoid and smoothness

Let $X_1, \dots, X_n \in \mathcal{F}$ be local generators of $\mathcal{F}$.

A **path holonomy bi-submersion** is (U, s, t) where

$$U \subset M \times \mathbb{R}^n \begin{matrix} \xrightarrow{s} \\ \xrightarrow{t} \end{matrix} M$$

and the (source and target) maps are

$$s(y, \xi) = y$$

$$t(y, \xi) = \exp_y(\sum_{i=1}^n \xi_i X_i), \text{ the time-1 flow of } \sum_{i=1}^n \xi_i X_i \text{ starting at } y.$$

There is a notion of composition and inversion of path holonomy bi-submersions, as well as a notion of morphism.

Take a family of path holonomy bi-submersions $\{U_i\}_{i \in I}$ covering M . Let $\mathcal{U}$ be the family of all finite products of elements of $\{U_i\}_{i \in I}$ and of their inverses.

The **holonomy groupoid of the foliation** $\mathcal{F}$ [Androulidakis-Skandalis] is

$$H := \coprod_{U \in \mathcal{U}} U / \sim$$

where $u \in U \sim u' \in U'$ if there is a morphism of bi-submersions $f : U \rightarrow U'$ (defined near u) such that $f(u) = u'$.

H is a topological groupoid over M , usually not smooth.

Examples 1) Consider the action of S^1 on $M = \mathbb{R}^2$ by rotations. Then

$$H = S^1 \times \mathbb{R}^2 \rightrightarrows \mathbb{R}^2$$

(the transformation groupoid).

2) Consider the action of $GL(2, \mathbb{R})$ on $M = \mathbb{R}^2$ and the induced foliation. Then

$$H = (\mathbb{R}^2 - \{0\}) \times (\mathbb{R}^2 - \{0\}) \coprod GL(2, \mathbb{R}).$$

3) Let $G \rightrightarrows M$ be a Lie groupoid with Lie algebroid B , and $\mathcal{F} = \rho(\Gamma_c(B))$. Then H is a quotient of G .

Smoothness of H_L

Let L be a leaf and $x \in L$. There is a short exact sequence of vector spaces

$$0 \rightarrow \mathfrak{g}_x \rightarrow (\mathcal{F}/I_x\mathcal{F}) \xrightarrow{ev_x} T_x L \rightarrow 0$$

where ev_x is evaluation at x .

$$A_L := \cup_{x \in L} (\mathcal{F}/I_x\mathcal{F})$$

is a transitive Lie algebroid over L , with $\Gamma_c(A_L) \cong \mathcal{F}/I_L\mathcal{F}$.

Question: When does A_L integrate to H_L (the restriction of the holonomy groupoid to L)?

Lemma

Let G_x be the connected and simply connected Lie group integrating the Lie algebra $\mathfrak{g}_x$. There exists a canonical morphism of topological groups

$$\varepsilon: G_x \rightarrow H_x^x.$$

ε is defined using bi-submersions.

$\text{Ker}(\varepsilon)$ is called **essential isotropy group** of the leaf L .

Theorem

Let $(M, \mathcal{F})$ be a foliation and L a leaf.

The transitive groupoid H_L is smooth and integrates the Lie algebroid A_L if and only if $\text{Ker}(\varepsilon)$ is discrete.

Open question: Is $\text{Ker}(\varepsilon)$ always discrete?

B) Holonomy

For a regular foliation F and a path γ in a leaf, the holonomy of γ is defined “following nearby paths in the leaves of F ”.

For singular foliations this fails (think of $M = \mathbb{R}^2$, $\mathcal{F} = \langle x\partial_y - y\partial_x \rangle$, and γ the constant path at the origin). •

Question: How to extend the notion of holonomy to singular foliations?

Let $x, y \in (M, \mathcal{F})$ be points in the same leaf L , and fix transversals S_x and S_y .

Theorem

There is a well defined map

$$\Phi_x^y: H_x^y \rightarrow \frac{\text{GermAut}_{\mathcal{F}}(S_x, S_y)}{\exp(I_x \mathcal{F})|_{S_x}}, \quad h \mapsto \langle \tau \rangle.$$

Here τ is defined as follows, given $h \in H_x^y$:

- *take any bi-submersion $(U, \mathbf{t}, \mathbf{s})$ and $u \in U$ satisfying $[u] = h$,*
 - *take any section $\bar{b}: S_x \rightarrow U$ through u of $\mathbf{s}$ such that $(\mathbf{t} \circ \bar{b})(S_x) \subset S_y$,*
- and define $\tau := \mathbf{t} \circ \bar{b}: S_x \rightarrow S_y$. •*

Example:

Let $M = \mathbb{R}$ and $\mathcal{F} = \langle x\partial_x \rangle$. We have $H = \mathbb{R} \times M \rightrightarrows M$.

So $H_0^0 \cong \mathbb{R}$, and a transversal S_0 at 0 is a neighborhood of 0 in M . We have:

$$\Phi_0^0(\lambda) = [y \mapsto e^\lambda y] \in \frac{\text{GermAut}_{\mathcal{F}}(S_0, S_0)}{\exp(I_0 x \partial_x)}.$$

We obtain a groupoid morphism

$$\boxed{\Phi: H \rightarrow \cup_{x,y} \frac{\text{GermAut}_{\mathcal{F}}(S_x, S_y)}{\exp(I_x \mathcal{F})|_{S_x}}.}$$

Conjecture: Φ is injective.

(Proven at points x where $\mathcal{F}$ vanishes, and for regular foliations.)

Remark: If $\mathcal{F}$ is a regular foliation, then $\exp(I_x \mathcal{F})|_{S_x} = \{Id_{S_x}\}$, hence the map Φ recovers the usual notion of holonomy for regular foliations.

The above conjecture and remark are two justifications for calling H *holonomy groupoid*.

Linear holonomy

Let L be a leaf, assume that $Ker(\varepsilon)$ is discrete.
From the holonomy map Φ we obtain:

1) by taking the derivative of τ :

$$\Psi_L: H_L \rightarrow Iso(NL, NL),$$

a Lie groupoid representation of H_L on NL .

2) by differentiating Ψ_L :

$$\nabla^{L,\perp}: A_L \rightarrow Der(NL),$$

the Lie algebroid representation of A_L on NL induced by the Lie bracket.

(Notice that $\Gamma(A_L) = \mathcal{F}/I_L\mathcal{F}$ and $\Gamma(NL) = \mathcal{X}(M)/(\mathcal{F} + I_L\mathcal{X}(M))$.)
Here $\Gamma(Der(NL)) = \{\text{first order differential operators on } NL\}$.

C) Linearization

Vector field Y on M tangent to $L \rightsquigarrow$
vector field Y_{lin} on NL , defined as follows:

Y_{lin} acts on the fiberwise constant functions as $Y|_L$

Y_{lin} acts on $C_{lin}^\infty(NL) \cong I_L/I_L^2$ as $Y_{lin}[f] := [Y(f)]$. •

The **linearization of $\mathcal{F}$ at L** is the foliation $\mathcal{F}_{lin}$ on NL generated by $\{Y_{lin} : Y \in \mathcal{F}\}$.

Lemma

Let L be an embedded leaf such that $Ker(\varepsilon)$ is discrete.

Then the linearized foliation $\mathcal{F}_{lin}$ is the foliation induced by the Lie groupoid action Ψ_L of H_L on NL .

We say $\mathcal{F}$ is **linearizable at L** if there is a diffeomorphism mapping $\mathcal{F}$ to $\mathcal{F}_{lin}$.

Remark: When $\mathcal{F} = \langle X \rangle$ with X vanishing at $L = \{x\}$, linearizability of $\mathcal{F}$ means: there is a diffeomorphism taking X to a fX_{lin} for a non-vanishing function f . It is a weaker condition than the linearizability of the vector field X !

Question: When is a singular foliation isomorphic to its linearization?
We don't know yet, but:

Proposition

Let L be an embedded leaf s.t. $Ker(\varepsilon)$ is discrete. Assume that H_x^x is compact for $x \in L$. The following are equivalent:

- 1) $\mathcal{F}$ is linearizable about L*
- 2) there exists a tubular neighborhood U of L and a (Hausdorff) Lie groupoid $G \rightrightarrows U$, proper at x , inducing the foliation $\mathcal{F}|_U$.*

In that case:

- G can be chosen to be the transformation groupoid of the action Ψ_L of H_L on NL ,
- $(U, \mathcal{F}|_U)$ admits the structure of a singular Riemannian foliation.

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Thank you!